## Components Required

| Component | Quantity | Description |
|---|---|---|
| QA250 drone frame | 1 | Carbon fibre drone frame |
| CC3D flight controller | 1 | Flight controller |
| 3C11.1V Lipo battery | 1 | 3300mAh battery |
| FlySky 6T6B radio receiver and transmitter | 1 | RC transmitter and receiver |
| 6046 propellers | 4 | Two each for CW and CCW |
| 2200kV motors, 6T | 4 | A2212 motor |
| 30/40A ESC | 4 | Simonk ESC |
| Power distribution board | 1 | Small power distribution board for drone |
| XT60 connector | 1 | Male female |
| AA alkaline battery | 8 | 1.5V battery |
| Raspberry Pi Zero W with camera | 1 | Raspberry Pi Zero W |
| AIY voice bonnet with speaker | 1 | Speaker |

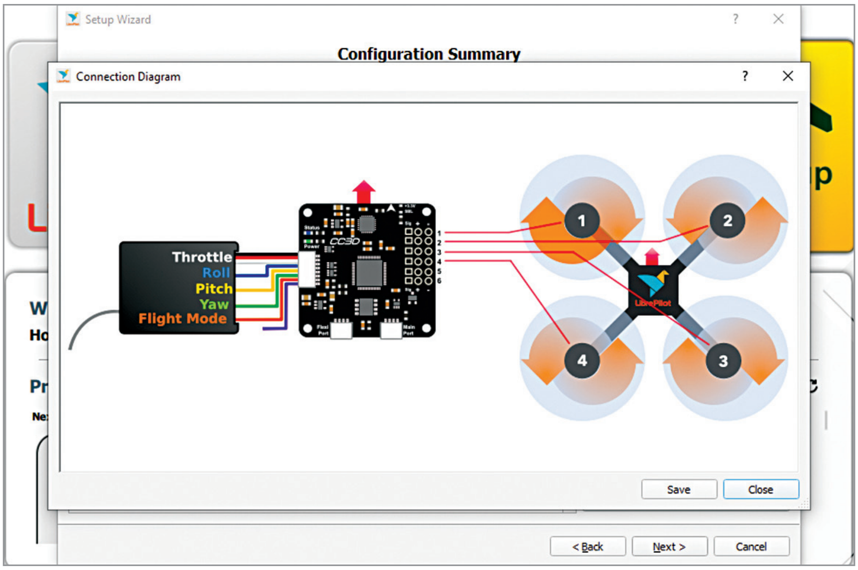


*Fig. 5: Pin diagram of CC3D flight controller*

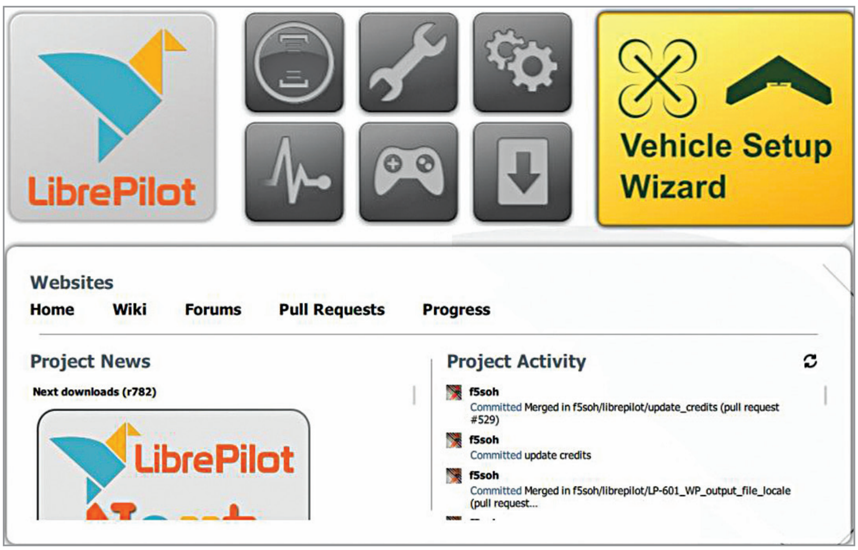


*Fig. 6: Vehicle Setup Wizard window in LibrePilot*

marked on the flight controller board. Black wire of ESC is ground (GND) pin that needs to be connected to '–' , red wire is positive pin that should be connected to ' + ', and yellow wire is PWM signal pin that may be connected to 'S' pin of the flight controller, respectively (refer Fig. 5). Next, connect the radio receiver to CC3D flight controller, as shown in Fig. 5.

### Preparing the flight controller

The flight controller can be programmed using OpenPilot ground control station (GCS) or LibrePilot GCS software. You can download and use any one of these. We used LibrePilot while testing.

Open LibrePilot software and connect the flight controller to a PC via a USB. From the LibrePilot software select Vehicle Setup Wizard (refer Fig. 6). From the range of options, choose to update the firmware by clicking on Upgrade and then wait for the firmware to upload on the flight controller. You will get a window (refer Fig. 7) where you can do ACC calibration.

Put the flight controller on a flat surface and enure it is facing towards the front motor. Calibrate the accelerometer and the motor ESC as per the given instructions. Disconnect the battery from the power distribution board. Then click on start and wait for a twitching/beeping sound from the motor. Repeat the procedure once again. The ESC is now calibrated, as shown in Fig. 8.

You are then taken to output calibration, as shown in Fig. 9. Here the rotational direction of a motor is shown. Click on start and then move the sliders until a motor starts moving smoothly. Then click stop. Ensure that the rotational direction of a motor matches with that shown on the screen of LibrePilot software. If not, then interchange any two wirings of the motor to ESC to change the rotational direction. Repeat the procedure for all motors and save the settings.

Next, it will take you to the transmitter receiver setup wizard. Make sure that the transmitter remote is on, the receiver is connected, and the LiPo battery is connected to the power distribution board. Follow the instructions given by the software.